



Integrating knowledge representation into traffic prediction: a spatial–temporal graph neural network with adaptive fusion features

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Abstract

Various external factors that interfere with traffic flow, such as weather conditions, traffic accidents, incidents, and Points of Interest (POIs), need to be considered in performing traffic forecasting tasks. However, the current research methods encounter difficulties in effectively incorporating these factors with traffic characteristics and efficiently updating them, which leads to a lack of dynamics and interpretability. Moreover, capturing temporal dependence and spatial dependence separately and sequentially can result in issues, such as information loss and model errors. To address these challenges, we present a Knowledge Representation learning-actuated spatial–temporal graph neural network (KR-STGNN) for traffic flow prediction. We combine the knowledge embedding with the traffic features via Gated Feature Fusion Module (GFFM), and dynamically update the traffic features adaptively according to the importance of external factors. To conduct the co-capture of spatial–temporal dependencies, we subsequently propose a spatial–temporal feature synchronous capture module (ST-FSCM) combining dilation causal convolution with GRU. Experimental results on a real-world traffic data set demonstrate that KR-STGNN has superior forecasting performances over diverse prediction horizons, especially for short-term prediction. The ablation and perturbation analysis experiments further validate the effectiveness and robustness of the designed method.

Keywords Traffic flow prediction · Spatial–temporal graph neural network · Knowledge representation learning · External knowledge fusion

Introduction

In recent years, with the continuous development of urban public transportation, motor vehicle ownership continues growing rapidly, increasing traffic congestion in large- and medium-sized cities [1]. How to alleviate urban traffic congestion, so as to reasonably dispatch urban traffic resources and improve urban traffic efficiency is an urgent problem

to be solved. With the continuous promotion of big data, artificial intelligence, and urban management tools, coupled with the integration of IoT systems, the field is gradually transitioning from digital to intelligent. The integration of IoT systems enhances the capability of collecting real-time data from multiple sensors and devices, which contributes to effective monitoring and analysis of urban environments. By incorporating IoT technology, the transition towards intelligent urban management becomes more seamless and efficient [2].

Accurate traffic flow forecasting can provide the basis for traffic management decisions and offer convenient services to motorists and travelers. Traffic state can be affected by many external factors. As shown in Fig. 1a, weather conditions can affect the traffic flow in the region, and different types of POIs around nodes in the traffic network or unexpected events and traffic incidents in the surrounding area can lead to different traffic states in the region. In addition, as shown in Fig. 1b, severe weather can affect road traffic conditions, and the traffic conditions between different road sections then have dynamic temporal and spatial correlations.

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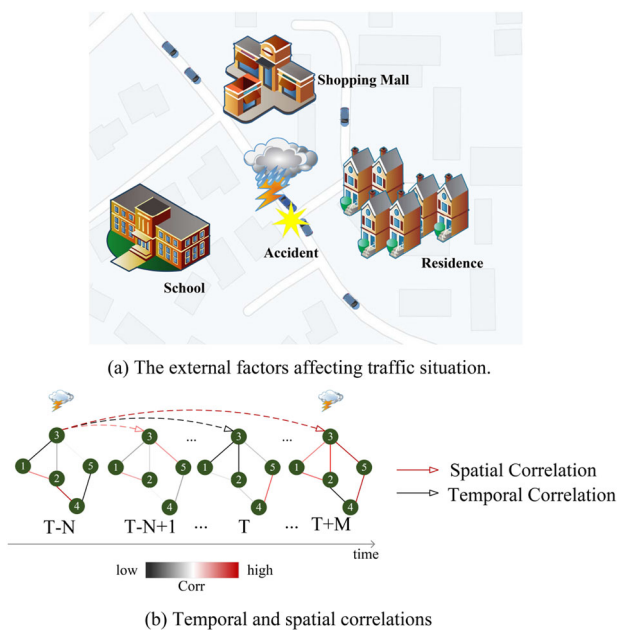


Fig. 1 External factors affecting the traffic situation, and temporal and spatial correlations. **a** Static factors include road attributes and POIs (e.g., shopping malls, schools). Dynamic factors include weather conditions and traffic flow. **b** Temporal correlation: The correlation of road section 3 increases between $T-N$ slice and $T+M$ slice due to thunderstorm. Spatial correlation: the correlation between road section 1 and road section 3 at $T-N$ slice differs from that at $T-N+1$ slice

With the continuous development of intelligent transportation systems, in addition to relying on traffic data captured by coils, intelligent traffic cameras, and other devices laid on the urban roads, more and more data affecting traffic flow can be collected. However, existing traffic flow prediction methods generally consider capturing the temporal and spatial characteristics from traffic flow data to predict future traffic flows, ignoring the influence of external factors on the effectiveness of traffic flow prediction. Therefore, how to consider external factors and combine them with traffic flow data to obtain superior traffic flow prediction results and improve prediction accuracy is a challenge that needs to be focused on.

The spatial propagation of the past traffic state determines the current traffic state of the road network. The existing domestic and international methods for traffic flow prediction can be divided into two main strategies: model-driven and data-driven approaches [3]. With the rapid development of IoT in transportation and big data technology, as well as the development of deep learning in the field of graph networks, researchers mainly study data-driven methods, especially from the perspective of spatial–temporal characteristics to study traffic prediction and other areas. Recurrent neural networks such as RNN [4], LSTM [5], and GRU [6] are used in time-dependent modeling due to their ability to capture time-series data. Graph neural networks are used to handle the capture of spatial dependence in traffic flow prediction

due to their ability to model non-Euclidean spatial data. The primary concept behind spatial–temporal graph neural networks is to integrate deep learning-based time series models effectively with graph neural networks, enabling a powerful and synergistic combination of these two approaches [7]. This integration has become the most popular method in traffic flow forecasting due to its ability to capture both spatial and temporal dependencies. If the influence of external factors is not considered, the capture of direct or indirect relationships between external factors and traffic information affecting urban traffic conditions will be affected. For example, severe weather conditions or the number of POIs around a road section during a specific period can affect the traffic flow in that area. Consequently, integrating multiple data sources with traffic features is of great importance for the forecasting task.

In addition to the sensor data obtained from the urban traffic system and the data obtained from social media, each area of the city contains several concepts that humans can understand within a package of hidden and inherent knowledge. Most current solutions to urban application problems rely only on deep learning approaches, which do not effectively use semantic information between various types of knowledge to prediction tasks and lack interpretation of prediction results. Knowledge graph technology can efficiently compute complex semantic information between entities and relations, which helps to capture macroscopic deep-level information and is easy to integrate with deep learning models [8]. Therefore, knowledge graph technology can provide rich knowledge and more diversified information for applications, such as traffic situation analysis and prediction. Furthermore, we can extract target information from traffic scenes in other fields, such as object detection [9]. This target information can be used as one of the traffic features to further improve prediction accuracy. In addition, the control strategy in the traffic flow prediction model can be optimized through optimal iterative learning control approach [10] and fixed-time prescribed performance trajectory tracking control [11] to improve the real-time and robustness of the model.

Some researchers have predicted traffic flow using knowledge of external factors as auxiliary information. Still, the external factors are usually directly encoded as representation vectors and then simply concatenated with traffic feature vectors. However, these methods do not integrate the relationships between external factors and traffic flow well, and then perform the representation of their correlation. In addition, because different external factors may have varying degrees of impact on traffic flow and congestion, our designed model can distinguish their respective importance to better understand the dynamic changes in the traffic system and incorporate them into the traffic prediction model to improve accuracy. Not only that, our designed traffic prediction model has stronger comprehensiveness and more abundant infor-

mation interaction, as it can simultaneously consider time dependence and spatial dependence, integrating them into a unified module enabling information interaction and transmission in both temporal and spatial dimensions. This allows for a more comprehensive analysis and capture of the spatio-temporal relationships in traffic data, further enhancing the model's expressive power. To reveal the interaction mechanism between traffic flow conditions of road networks and external factors, we propose a spatial–temporal graph neural network method based on knowledge representation learning to predict traffic flow. This method considers the fusion of multi-source data and makes the traffic information and external factors together as the input of the heterogeneous semantic network representation by constructing a knowledge graph, and then adopts the knowledge representation learning (KRL) method to obtain the semantic information between them. KRL methods can effectively encode these complex external factors and express them as low-dimensional vectors. By combining these vectors with traffic features and inputting them into the prediction model, the accuracy and robustness of the model can be improved, making the traffic prediction results more accurate and reliable. In addition, GFFM is proposed and the updated features through it are introduced into the spatial–temporal graph convolutional network, and the spatial–temporal dependencies are captured by the ST-FSCM. We utilize KR-STGNN for traffic prediction based on the Shenzhen taxi data set, and there is a corresponding improvement in each evaluation index compared to baseline models. The main contributions of this paper can be summarized as follows:

1. We propose a knowledge-actuated model for traffic flow prediction, where to utilize external knowledge to update traffic information, and we design GFFM to adaptively integrate external knowledge with traffic features according to their importance to better understand the dynamic changes of the traffic system.
2. We propose an ST-FSCM to realize the co-capture of temporal and spatial level features, which further reveals the essence of spatial–temporal correlation, and realizes the interaction and transmission of information between the temporal and spatial dimensions.
3. We evaluate our approach using a real-world traffic data set. The results show that KR-STGNN possesses a lower prediction error compared with the baseline methods, proving the superiority of the model in traffic prediction.

Relate work

Traffic flow prediction

Traffic flow prediction is one of the critical tasks in intelligent transportation systems. Typical traffic flow prediction

methods are divided into model-driven methods and data-driven methods. Model-driven methods use mathematical tools, such as physical knowledge and differential equations, rely on dynamic modeling, and formulate traffic problems through computational simulation [12]. Examples include traffic speed models and queuing theoretical models. However, these models may not account for external factors affecting the actual traffic situation, making it challenging to simulate real-world traffic effectively. With the increasing number of devices generating heterogeneous traffic data and the growing complexity of traffic patterns, data-driven methods have become increasingly predominant. Data-driven approaches rely on statistical data patterns to predict traffic trends [13].

Since then, two major data-driven predictive models have emerged: statistical analysis and machine learning. The advantages of statistical analysis methods are simple algorithms and easy implementation. However, due to the highly dynamic and nonlinear nature of traffic data, these methods cannot be able to effectively and accurately analyze future traffic conditions. Machine learning models can be used for nonlinear data model building by extracting traffic-related features and training the models while reducing prediction errors [14, 15].

However, traditional machine learning models are only applicable to scenarios with simple traffic conditions or small traffic data size because of their simple model architecture and limited parameters, and low computational efficiency when capturing temporal features. With the continuous rise of deep learning, feedforward neural networks, deep belief networks, and RNN are widely used in traffic prediction tasks with various complex relationships. RNN can learn the long-term temporal correlation of traffic data, and its variants LSTM, as well as GRU, can further enhance the ability of RNN to model remote time-series traffic data [16]. However, these methods only model single-node traffic time series data, ignoring the effect of spatial dependence on traffic conditions at each node. Convolutional neural network (CNN) was introduced to traffic prediction tasks to extract the spatial features of road networks. Still, it can only handle data in the Euclidean domain, and traffic road networks belong to data structures in non-Euclidean space, so this method is limited to the traffic domain. Based on this, graph neural network (GNN) was proposed to solve the non-Euclidean structure problem by the adjacency matrix between the nodes of the road network. A similar approach can be utilized in the field of wireless networks to model and predict the state and performance of wireless networks to improve problems in network resource allocation, interference management and user experience [17]. The introduction of GNN can better capture the topology and dynamic changes between nodes in a wireless network and learn the feature information and

relationships of nodes for state prediction and performance optimization.

Graph convolutional network (GCN) is suitable for any topology because of its ability to learn both node feature information and structure information [18]. STGCN [19] utilizes convolutional networks to efficiently extract valuable temporal features from complex traffic data with a graph structure, while Graph Wavenet [20] employs sophisticated dilated causal convolutions to effectively capture and model the intricate temporal dependencies within the data. T-GCN [21] combines GCN and GRU to learn complex topologies to obtain spatial correlation and dynamic changes of traffic data to get temporal correlation.

In addition to historical traffic information, traffic conditions can be affected by various external factors, such as weather conditions (rainfall, temperature, humidity, air quality), bus stations, POIs, special events, and other factors when performing forecasting tasks. Some factors, such as weather conditions and date attributes, are encoded as discrete values, which are generally converted to binary vectors using one-hot encoding. Temperature, humidity, and wind speed are encoded as continuous values and processed using Min–Max normalization method or Z-score normalization method. The main challenge of current traffic prediction tasks is to integrate the information of external factors into the prediction model. The summary of traffic flow prediction methods based on considering external factors is shown in Table 1. Several methods exist in existing studies to deal with external factors, one approach is to concatenate external factor features with other features and input them into the model. Xie et al. [24] proposed a deep incident-aware graph convolutional network (DIGC-Net), which effectively combines urban traffic events, spatial–temporal, periodic, and contextual features for traffic

speed prediction. Furthermore, researchers have explored the use of recurrent neural networks to incorporate external factors. Hou et al. [30] design external attribute augmentation units to extract external factors affecting traffic flow, which are then input to the Leigh LSTM to learn the complex spatio-temporal dependence and external dependence of traffic data. Another approach is to design a component dedicated to handling external factors, some of which utilize KRL methods. Long et al. [33] designed a gated fusion module to reduce prediction errors by fusing external factors with spatio-temporal features in front of the traffic prediction model. Zhu et al. [34] proposed an attribute-augmented spatial–temporal graph convolutional network in which external factors are modeled as dynamic and static attributes, and an attribute enhancement unit is designed for these factors to be encoded and integrated into the spatial–temporal graph convolutional model. Wang et al. [35] proposed a KRL-based method for subway traffic flow prediction. The influence of external factors is incorporated into the prediction model based on spatio-temporal graph neural network by constructing a knowledge graph that stores the factors related to the subway traffic network and learning the influence representation of external factors based on KRL technique. Xiong et al. [36] similarly use KRL to quantify external heterogeneous information and thus use the heterogeneity of this attribute information for traffic flow prediction.

To summarize, among the various methods for handling external factors, concatenation methods, recurrent neural network methods, and gate fusion methods suffer from feature space expansion and increased computational complexity, in addition to being susceptible to long-term dependency problems and the possibility of losing some important information during the fusion process. In contrast, the KRL method

Table 1 Summary of traffic flow prediction methods based on considering external factors (part)

Methods	Directions	Types of External Factors	Approaches
GTCN [22]	Traffic Speed Prediction	Weather, Holiday	Concatenate
GCN–DHSTNet [23]	Traffic Flow Prediction	Event, Holiday, Weather	Concatenate
DIGC-Net [24]	Traffic Speed Prediction	Traffic incident, Weather	Concatenate
STNN [25]	Traffic Speed Prediction	POI, Road feature	Concatenate
MVGCN [26]	Traffic Flow Prediction	POI, Road feature	Concatenate
CACLSTM [27]	Traffic Flow Prediction	Weather, Holiday	Concatenate
AST–GCN [28]	Traffic Speed Prediction	Weather, POI	Concatenate
Attention–Seq2seq [29]	Traffic Flow Prediction	Metadata	LSTM
AST–GCN–LSTM [30]	Traffic Flow Prediction	Weather, POI	LSTM
CNN–LSTM [31]	Traffic Flow Prediction	Weather	LSTM
[32]	Traffic Flow Prediction	Temperature, Weather	GRU
USTAN [33]	Traffic Flow Prediction	Weather, POI	Gated fusion
KST–GCN [34]	Traffic Speed Prediction	Weather, POI	KRL
KGR–STGNN [35]	Traffic Flow Prediction	Weather, Social media	KRL
HIT–GCN [36]	Traffic Speed Prediction	Weather, POI	KRL

has its advantages, which not only provides better feature extraction capability and enhances the interpretability, but also reduces the dependence on a large amount of training data, thereby effectively quantifying the influence of external heterogeneous information and improving the prediction performance.

External multi-source data fusion

The mining and fusion of urban multi-source data is an important task in the construction of smart cities. With the rapid development of Internet and Internet of Things technology, the data generated in cities presents the characteristics of diversification and heterogeneity. These data include Internet web data, taxi sensor data, monitoring data, etc [37]. Furthermore, heterogeneous data types include text data, numerical data, video data, etc. How to make full use of these multi-source heterogeneous data, extract, integrate, and store it has become an urgent research problem.

To solve this problem, knowledge graphs are widely used to integrate multiple data sources to build large-scale knowledge bases. Especially in the field of transportation, the knowledge graph can store multi-source heterogeneous data, which has greater advantages than traditional relational databases. The knowledge graph stores data in the form of a graph structure and is very sensitive to the relationship between data, which makes it easier to mine the potential relationship between things [38]. Therefore, the combination of knowledge graph and deep learning can effectively improve the ability to interpret and utilize urban multi-source heterogeneous data. In the field of transportation, it is of great significance to integrate multi-source traffic data to construct a traffic knowledge graph. The traffic knowledge graph can contain rich spatial knowledge, such as road section information, relationship attribute information between road sections, and POIs. Through the mining of these multi-source data, more comprehensive and accurate spatial relationships can be obtained, and more diversified information can be provided for traffic forecasting and planning. In addition, using the spatial–temporal graph neural network to analyze and predict on the knowledge graph can better capture the temporal and spatial dependencies in traffic data, thereby improving the accuracy of traffic prediction.

In summary, the integration and analysis of diverse and heterogeneous urban multi-source data play a vital role in the advancement of smart city development. By combining deep learning techniques with knowledge graphs, we can effectively interpret and harness the potential of this data. Specifically, incorporating various sources of traffic data into a traffic knowledge graph allows us to capture comprehensive spatial relationships. As a result, integrating multi-source

traffic data into a traffic knowledge graph not only enables the comprehensive understanding of spatial relationships but also enhances traffic prediction accuracy through the utilization of spatial–temporal graph neural networks.

Methodology

Problem definition

In this research, the purpose of traffic flow prediction is to predict the traffic information in the future period by the traffic information captured by the road sensors in a certain historical time period. The main feature parameters of traffic state generally include traffic flow, traffic speed, and occupancy. In the experimental part of this study, traffic speed is chosen as the traffic node feature.

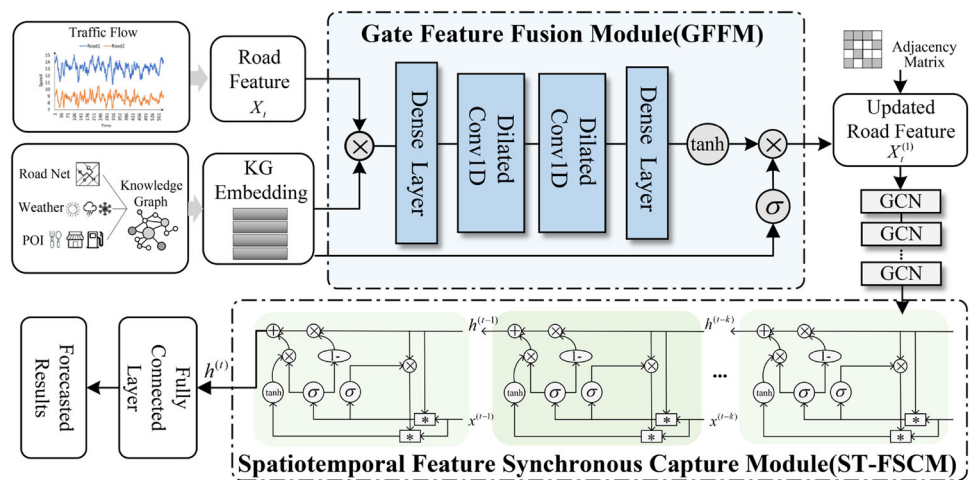
We utilize the traffic knowledge graph G to characterize the topology of the road network incorporating external factors. We consider each road as a node, and the traffic prediction problem can be viewed as calculating the traffic features for a future period by learning a function f based on the traffic network topology A , the feature matrix X and the traffic knowledge graph G :

$$y = f(X, A, G), \quad (1)$$

where X represents the features of each node in the urban road network. For the feature matrix X , we set it as the attribute features of traffic information in the nodes of the road network, denoted as $X \in \mathbb{R}^{N \times H}$, where H represents the number of node attribute features, N is the number of roads, and $X_t \in \mathbb{R}^{N \times t}$ represents the speed information on each road at time t . $A \in \mathbb{R}^{N \times N}$ is the adjacency matrix of the road network, the matrix contains only elements 0 and 1. If the corresponding road section i and road section j are connected, the element is 1, otherwise it is 0.

To consider the correlation between traffic information and external factors when predicting traffic flow, we first construct a traffic knowledge graph, then use the knowledge graph representation learning model to embed the semantic information in the knowledge graph into a low-dimensional space. To combine the external knowledge with the spatial–temporal graph convolutional network, GFFM is designed, which integrates and updates the embedding vectors and traffic features as its input and feeds them into the spatial–temporal graph convolutional network. Through this method, KR-STGNN can capture the spatial–temporal features of traffic in addition to the knowledge structure and semantic information between traffic information and external factors. The workflow of KR-STGNN is shown in Fig. 2.

Fig. 2 KR-STGNN framework



Traffic knowledge graph

Traffic knowledge graph construction

Knowledge graph is generally represented by entities, relationships, and attributes, where entities represent something that exists in the physical world, corresponding to nodes in the knowledge graph [39]. Relationships are the external links between things and things used to interconnect many nodes. As the representation of knowledge graph, triples are composed of entities and relations or entities, attributes, and attribute values. Attributes are embodied within entities or relationships and are inherent characteristics of entities. Attribute values are the characteristics, features, or parameters of attributes. How to capture the road network structure information of the spatial graph and apply it into downstream applications, it is necessary to construct the geographic spatial graph composed of traffic equipment, fixed POIs and road network structure, namely, the urban traffic knowledge graph. As shown in Fig. 3, it is visually displayed in the form of a knowledge graph, which corresponds to the “knowledge graph” part in the upper left corner of Fig. 2. The purpose of this is to display traffic information more intuitively and comprehensively, and to help us better understand and analyze the influencing factors in the transportation network.

The urban traffic knowledge graph can be represented in the form of a triple $G = \{E, R, T\}$, where E denotes the set of knowledge entities, R denotes the set of association relationships, and T denotes the set of triples. The triples involved in this paper include entity-relationship triples and entity-attribute triples, where entity-relationship triples can be represented by $\{entity1, relationship, entity2\}$ and entity-attribute triples can be represented by $\{entity, attribute, attribute\}$. As shown in Fig. 3, $\{Shenzhen, belongs\ to, Luohu\ District\}$ indicates that Luohu District belongs to Shenzhen, $\{Road1, adjacent, Road2\}$ indicates that Road1 is adjacent to Road2, $\{Road1, Catering\ Service, 4\}$ indi-

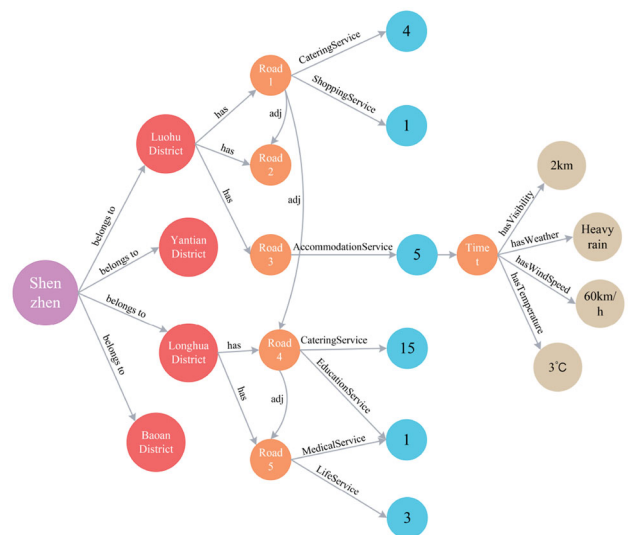


Fig. 3 Example of Shenzhen traffic knowledge graph

cates that there are four catering service POIs around Road1, $\{99903, 2015/1/4\ 20:30, fog\}$ indicates that the weather condition of the road section with ID 99903 is fog at 20:30 on January 4, 2015.

Knowledge representation learning

Knowledge Representation Learning is the study of how to effectively and efficiently extract and utilize helpful information from large-scale knowledge graphs for downstream prediction tasks [40]. The method based on embedding representation can not only represent knowledge well but also have satisfactory execution efficiency, and can map the implicit semantic information in the graph structure to the Euclidean space, so that the previously difficult to observe relationship becomes more obvious [41]. The existing techniques for knowledge representation learning can be categorized

into rule-based methods, embedding-based methods, and neural network-based methods. Among these, the neural network-based methods can be extended utilizing attention mechanism [42] and other techniques. However, for the task of traffic prediction, to consider the issues of computational efficiency and flexibility, we adopt the embedding representation-based method TransR [43] based on the translation model.

Figure 4 illustrates the fundamental principle of the TransR model. The key concept of this approach is to create distinct spaces for entities and relationships, and employ a transfer matrix to map entities from the entity space to the relationship space, thereby achieving the semantic vectorization of triples [44, 45]. It regards the relation vector as a vector translation operation from the head entity t_1 to the tail entity t_2 , and r represents the relationship between entities. For each triple (t_1, r, t_2) , where the entities t_{1r} and t_{2r} located in the entity space and are projected into the relationship space through a projection matrix N_r . Using the mapping matrix, the projection vector of the entity can be defined as

$$\begin{aligned} t_{1r} &= t_1 N_r, \\ t_{2r} &= t_2 N_r. \end{aligned} \quad (2)$$

The corresponding score function for measuring the distance between two vectors is defined as follows where enforcing constraints on the embeddings t_1, t_2 and the norm of the mapping matrix can make the relevant parameters of the model smaller to avoid the problem of overfitting and make it have a strong generalization ability. $\|\cdot\|$ indicates that the vector distance is calculated as L_2 parametric:

$$f(t_1, t_2) = \|t_1 + r - t_2\|_2^2. \quad (3)$$

In addition, the larger the score value is for the correct triples, the smaller the score value is for the incorrect triples. The loss function of the TransR model is defined as follows, and the minimization of the loss function value is the training objective of the model:

$$L = \sum_{(t_1, r, t_2)} \sum_{(t'_1, r, t'_2)} [\gamma + f_r(t'_1, t'_2) - f_r(t_1, t_2)]. \quad (4)$$

To utilize the knowledge of external factors in the traffic flow prediction model, it is first necessary to pre-train the constructed traffic knowledge graph using a knowledge graph representation learning method, and then obtain a semantic representation of traffic that can express the influence of external factors. The input of the pre-trained model includes static road network information and POIs and dynamic weather information. The semantic representations of 156 road section entities in Luohu District of Shenzhen are obtained by pre-training, and the embedding dimension

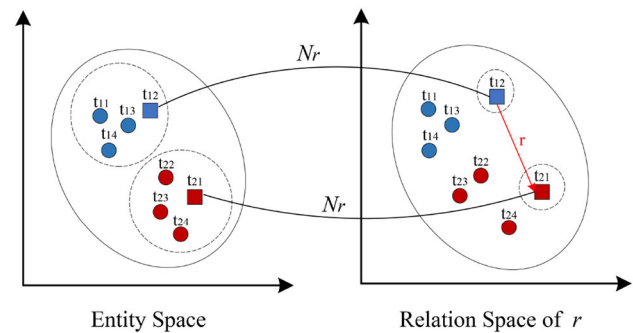


Fig. 4 Knowledge graph representation learning based on TransR model

of each semantic representation is set to 20. The formula for pre-training using the knowledge graph representation learning method is as follows:

$$X_{embedding} = TransR \{(h, r, t) \cup (h, att, att_val)\}, \quad (5)$$

where $X_{embedding}$ denotes the semantic representation of 156 traffic sections, (h, r, t) denotes the entity-relationship class triple, and (h, att, att_val) denotes the entity-attribute class triple.

To summarize, integrating knowledge graph into traffic prediction can enhance performance by abstracting traffic data and their interrelationships across multiple dimensions into nodes and edges, creating a unified data structure. This allows the model to better comprehend data correlations and adapt to complex environmental changes and noise. Furthermore, knowledge representation encodes semantic relationships between various dimensions of data, maximizing the utilization of feature semantics and further improving the model's robustness and predictive accuracy.

Gated feature fusion module

A common approach to capturing the impact of external factors on urban traffic is to directly concatenate external factors with traffic features. However, it is not accurate to reflect the impact of external factors on spatial-temporal networks by directly concatenating external factors with traffic features with the same weights. Based on this, Gate Feature Fusion Module (GFFM) is designed to update and integrate different external factors and traffic features adaptively and dynamically according to the importance of each component. This module can be applied to traffic prediction models based on graph neural networks.

According to the diversity of external factors, we classify the external factors into static factors (POI) and dynamic factors (weather conditions). The entity and relationship semantic information of static and dynamic factors are mapped to the vector space by the KRL method. The cor-

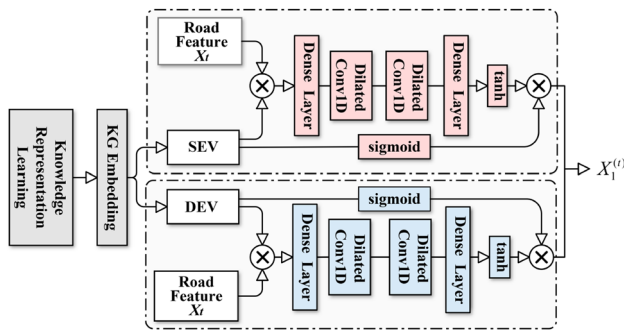


Fig. 5 Gate feature fusion module (GFFM)

responding embedding vector representations SEV (Static Embedding Vector) and DEV (Dynamic Embedding Vector) are obtained as X_{SEV} , X_{DEV} , respectively. Figure 5 shows the calculation process of the gated feature fusion module, where the module inputs are the embedding vectors X_{SEV} and X_{DEV} after the KRL model, and the traffic features X_t observed at time t . The feature extraction of traffic features is performed by one-dimensional dilation convolution [46]. This operation can expand the receptive field along the time axis, thereby enhancing the performance of the module in extracting sequence correlation.

$$\begin{aligned} X_{st} &= *_{DC} (X_{SEV} X_t) \tanh, \\ X_{dt} &= *_{DC} (X_{DEV} X_t) \tanh, \\ X_{sm} &= \sigma X_{SEV}, \\ X_{dm} &= \sigma X_{DEV}, \\ X_1^{(t)} &= [X_{st} X_{sm}, X_{dt} X_{dm}]. \end{aligned} \quad (6)$$

As shown in Eq. (6), where indicates one-dimensional dilation convolution, X_{sm} and X_{dm} indicate that the external factors are selected by sigmoid function for feature selection to obtain weight information. The output $X_1^{(t)}$ of the module represents the result of multiplying the weighted output of the two types of external factor features through the sigmoid activation function with the output through tanh, and thus combining them. This process in which the embedding vector X_{SEV} and X_{DEV} obtains the weight information through the activation function sigmoid aims to update it with the processed section features at the feature level. Since the traffic condition of a road section in a specific area is not only influenced by the previously observed data, but also by the distribution of POIs in the area. For example, if the weather condition of a road section is severe at a certain moment, such as heavy rain, or if the distribution of POIs in the area is relatively dense, then the impact of the captured external factors on the traffic flow is relatively large, and the cascade of traffic features and knowledge embedding will be larger. On the contrary, if the value of the captured external factors is smaller, the cascade of traffic features and knowledge

embedding is relatively small. This module can further enrich the feature diversity, further filtering and dynamic cascading of features, and extracting deeper feature dependencies, and the updated features are defined as $X_1^{(t)}$.

Spatial-temporal feature synchronous capture module

To construct a traffic prediction model based on knowledge representation, we use the updated road section features $X_1^{(t)}$ and adjacency matrix A as the input of the ST-FSCM. The input is first passed through graph convolution GCN, which is more suitable for non-Euclidean structured data in traffic situations compared to CNN, and the GCN captures the topological relationships and features of the road network using graph spectral theory to capture the representation vector of each road section as follows:

$$h'_{l+1} = \sigma(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} h'_l W_l), \quad (7)$$

where σ is the sigmoid activation function, $\tilde{A} = A + I_N$ is the adjacency matrix containing self-connections, I_N is the unit matrix, $\tilde{D}$ is the degree matrix of $\tilde{A}$, W_l denotes the trainable weight matrix parameter of the l th layer of convolution, and h'_l is the output of the l th layer of node features in the first layer of convolution, h'_l is $X_1^{(t)}$ after passing the gated feature fusion module:

$$h'_l = X_1^{(t)}. \quad (8)$$

To enable co-capture of spatial-temporal dependencies, we propose the dilated causal convolution gate recurrent unit, as shown in Fig. 6. GRU controls the use of historical information by updating gates and reset gates. The update gate is used to control the degree of input historical traffic state information at time t , and the reset gate is used to control the degree of combination of traffic information at time matrix t and historical traffic state information. As shown in Fig. 2, the spatial-temporal features of the road network are captured

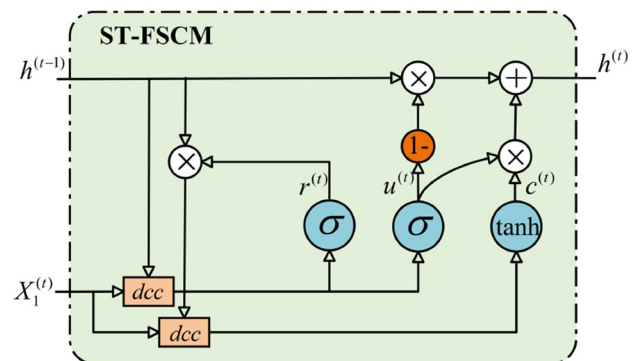


Fig. 6 Spatial-temporal feature synchronous capture module (ST-FSCM)

by propagating the information with node features after GCN into ST-FSCM. Causal convolution can solve the problem of different input and output time steps and future information leakage in the convolutional neural network model [47]. Furthermore, the causality and contextual relevance of the input time series are considered to capture the long-time information features by expanding the receptive field. It works, because the value at time t at each level is only jointly determined by the value at time t at the previous level and the value before time t , and is a structure oriented in time. Dilated convolution can preserve the temporal causal order without increasing the parameters and model complexity by expanding the receptive field by filling the input with zero values, so that only historical information is involved in predicting the current time step. Given a one-dimensional sequence input $x \in \mathbb{R}^T$ and a filter $g \in \mathbb{R}^K$, the dilated causal convolution of x and g at time t is represented as follows:

$$dcc(x, g) = \sum_{m=0}^{k-1} g(m)x(t - n \times m), \quad (9)$$

where n is the dilation factor used to control the jump distance and k is the convolution kernel size. Dilated causal convolution is calculated by first sampling the inputs at moment t and before moment t at intervals of size $n-1$, and then multiplying them with the convolution kernel $g(m)$ to calculate the inner product, so that the output is determined by the current input and the previous inputs. In addition, dilation causal convolution uses a dilation factor n to expand the perceptual field, which grows exponentially with the depth of the network, allowing the model to analyze early data effectively even with few network layers. Compared to models such as RNN and LSTM that capture the temporal correlation of the road network, GRU offers a simplified structure that reduces the number of parameters and computations required. In addition, GRU excels in managing memory by controlling the transfer of historical information through reset gates, leading to improved performance in addressing long-term dependency problems. Taking all these factors into consideration, we choose to integrate the ST-FSCM module with GRU to effectively capture spatiotemporal features in our traffic prediction task. We replace the fully connected operation in the traditional GRU with the dilated causal graph convolution operation, denoted as dcc in Fig. 6, to obtain ST-FSCM, through which the hidden state $h^{(t)}$ of the encoder at the current moment is obtained, which is computed as follows:

$$\begin{aligned} u^{(t)} &= \sigma(W_u \star_{dcc}([X_1^{(t)}, h^{(t-1)}], A) + b_u), \\ r^{(t)} &= \sigma(W_r \star_{dcc}([X_1^{(t)}, h^{(t-1)}], A) + b_r), \\ c^{(t)} &= \tanh(W_c \star_{dcc}([X_1^{(t)}, (r^{(t)} \odot h^{(t-1)})], A) + b_c), \\ h^{(t)} &= u^{(t)} \odot h^{(t-1)} + (1 - u^{(t)}) \odot c^{(t)}, \end{aligned} \quad (10)$$

where $\star_{dcc}$ is the dilated causal convolution defined in Eq. (9), $\odot$ denotes the Hadamard product, $r^{(t)}$ and $u^{(t)}$ are the reset gate and update gate for selecting or forgetting the memory, respectively, $\sigma(\cdot)$ and $\tanh(\cdot)$ are both activation functions, $X_1^{(t)}$ and $h^{(t)}$ are the traffic state information and output information at the current moment, respectively, $h^{(t-1)}$ denotes the hidden state information at the output at moment $t-1$, $c^{(t)}$ represents the traffic information stored in the memory cell at the current moment. W_u , W_r , W_c and b_u , b_r , b_c represent the weights and biases of update gate, reset gate, and memory unit during training, respectively. Finally, $h^{(t)}$ is output to the fully connected layer to generate the predicted future speed $\hat{Y}_T$. In the process of the spatial-temporal graph convolutional network model, the ultimate goal is to make the error between the real speed value in the road network and the predicted value achieve a relative minimum. The loss function of the model is shown below. Where we denote $\hat{Y}_T$ as the predicted value, Y_T as the real value of the traffic, λ as the regularization hyperparameter, and L_{reg} as the L_2 regularization term that is to prevent overfitting. Algorithm 1 presents the entire training process of the proposed model. The time complexity of our model is primarily influenced by the size of the knowledge graph, which consists of the number of entity nodes and the number of relationships. For instance, the greater the number of road nodes, the higher the complexity of the model:

$$Loss(Y, \hat{Y}_T) = \|Y_T - \hat{Y}_T\| + \lambda L_{reg}. \quad (11)$$

Algorithm 1 KR-STGNN Training Algorithm

Input: Traffic speed data of road feature in the graph G : $X_t \in \mathbb{R}^{N \times t}$, the time span of historical traffic data L , the number of entities N and the number of relationships E in knowledge graph.

- 1: Calculate embedding vector X_{SEV} and X_{DEV} based on static external factor triples and dynamic external factor triples, respectively, via Eq. (5);
- 2: Calculate updated feature $X_1^{(t)}$ by fusing X_{SEV} and X_{DEV} using GFFM, merging them with X_t via Eq. (6);
- 3: Utilize $X_1^{(t)}$ and adjacency matrix A as the inputs for GCN, employ $X_1^{(t)}$ as the initial value for h_1^t via Eq. (7);
- 4: Calculate y_t , $h^{(t)}$ via Eq. (10).

Output: Y_T represents the predicted value of traffic speed.

Experiments

Data set

In this subsection, we evaluate the performance of the proposed model using the traffic data SZ-taxi from Shenzhen city, China, based on a real public transportation data set.

1. SZ-taxi: this data set contains taxi trajectory data of 156 major roads in Luohu District, Shenzhen, for a total of

1 month from January 1, 2015 to January 31, 2015. The data includes two parts, one is traffic speed data, where each column represents the speed of each road section at a specific timestamp, and the data are collected at an interval of 15 min. The other is to describe the connectivity of the road network, where each row represents each road, and the 156×156 adjacency matrix is used for modeling.

2. SZ-POI: This data set represents the POI information around each road section, which includes nine categories, including catering services, businesses, shopping services, transportation facilities, education services, life services, medical services, accommodation services, and other services. By comparing the number of various POI types around each road section, the POI type with the largest proportion is selected as the feature of the road section, and a static matrix with the size of 156×2976 is constructed.
3. SZ-weather: This data set represents the weather state of each road section recorded every 15 min for 31 days, and a total of five weather states are recorded, which are sunny, cloudy, foggy, light rain, and heavy rain, and a dynamic matrix of size 2976×156 is constructed.

Experimental setting

The test platform settings used in the experiments are as follows: CPU: Intel(R) Core(TM) i9-11900K @ 3.50GHz, GPU: NVIDIA GeForce GTX 3090, Python3.7, Tensorflow1.14.0. During this experiment, we use the first 12 time steps as input to predict the traffic speed for the next 1, 2, 3, and 4 time steps, where the unit time step is 15 min. 80% of the data set is used as the training set and 20% as the test set. Adam is used as the optimizer to train our model. The learning rate is set to 0.001, the number of training epochs is 3000, the batch size is 64, and the number of hidden units of GRU capturing time dependence in ST-FSCM is set to 64.

Since different knowledge embedding dimensions affect the model performance, we need to choose the optimal embedding dimension as a kind of hyperparameter in the KRL process before the GFFM module, and we choose the embedding dimensions of 5, 10, 20, 30, 40, 50 and 60 for comparative analysis. Figure 7 shows that the performance of the model is relatively best when the embedding dimension is 20 by comparison of the evaluation indicators. This is mainly because the training data are not very large, and if the embedding dimension is larger than a certain level, it will lead to overfitting. Therefore, we fix the embedding dimension as 20 in this study for experiments.

Baseline methods

We compare the model KR-STGNN with the following baseline models:

- SVR [13]: Support Vector Regression model, which is a machine learning method for regression of traffic flow using support vector machines.
- ARIMA [48]: Autoregressive Integrated Moving Average, which is a classical time series forecasting model.
- DCRNN [49]: Diffusion Convolutional Recurrent Neural Network, which is one form is dilated convolutional recurrent neural network with encoder–decoder structure, which combines dilated graph convolutional network with GRU to capture the spatial–temporal dependencies of traffic data.
- STGCN [19]: Spatio-temporal Graph Convolutional Networks, which is a traffic flow prediction method using CNN to learn spatio-temporal features. In terms of spatial features, GCN is used to obtain the spatial dependencies between nodes, and one-dimensional convolution is used to model the traffic flow sequence in terms of temporal features.

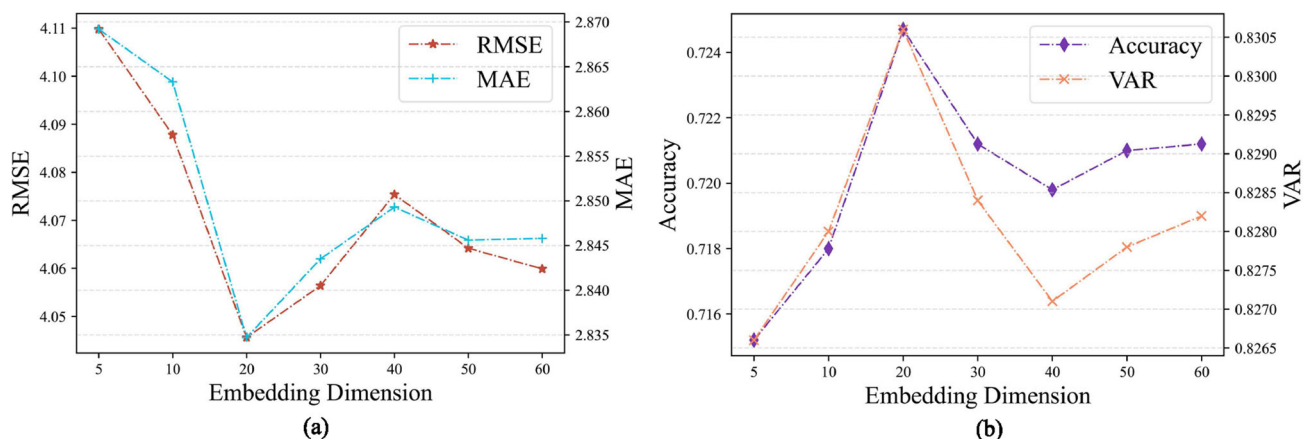


Fig. 7 Performance of KR-STGNN with different embedding dimensions. **a** RMSE and MAE. **b** Accuracy and VAR

- STSGCN [50]: Spatio-temporal Synchronous Graph Convolutional Networks, which is a GCN model that synchronously captures spatio-temporal dependences by building multiple local spatio-temporal graphs.
- KST-GCN [34]: Knowledge-Driven Spatio-temporal Graph Convolutional Network, which is a method that integrates external information and spatio-temporal information, and uses knowledge representation-driven method for traffic prediction on the basis of spatio-temporal graph convolutional network.
- AdapGL [51]: Graph convolutional network-based adaptive graph learning, which is an adaptive graph learning algorithm based on GCNs to capture complex dependencies.

We demonstrate the ability of our model to capture the nonlinear dependence of the traffic road network by comparing it with our model using traditional algorithms SVR and ARIMA. By comparing with DCRNN, STGCN, STSGCN and AdapGL, we demonstrate that updating traffic features using external factors and capturing spatial-temporal features together can effectively improve the prediction results. In addition, we make a comparison using KST-GCN to demonstrate that GFFM can effectively update traffic features.

Evaluation metrics

To verify the effectiveness of the model, we used five evaluation metrics to evaluate the performance of the model: Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Accuracy (ACC), Explained Variance Score (VAR), Coefficient of Determination (R^2). Where RMSE and MAE are used to measure the absolute error between the predicted and true values; ACC is used to measure the model prediction accuracy; VAR is used to measure the degree to which the model explains the fluctuations in the data set, and R^2 is used as a comprehensive measure of the goodness of fit of the regression model to the sample observations. Simultaneously, these evaluation metrics can complement each other to improve the overall assessment of model performance and help select and optimize models suitable for traffic prediction tasks. They are calculated as shown below:

- (1) Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{MN} \sum_{j=1}^N \sum_{m=1}^M (y_m^j - \widehat{y}_m^j)^2}.$$

- (2) Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{MN} \sum_{j=1}^N \sum_{m=1}^M |y_m^j - \widehat{y}_m^j|.$$

- (3) Accuracy (ACC):

$$\text{ACC} = 1 - \frac{\|y - \widehat{y}\|_F}{\|y\|_F}.$$

- (4) Explained Variance Score (VAR):

$$\text{VAR} = 1 - \frac{\text{Var}\{y - \widehat{y}\}}{\text{Var}\{y\}}.$$

- (5) Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{\sum_{j=1}^N \sum_{m=1}^M (y_m^j - \widehat{y}_m^j)^2}{\sum_{j=1}^N \sum_{m=1}^M (y_m^j - \bar{y})^2},$$

where y_m^j and $\widehat{y}_m^j$ represent the real value and predicted value of one of the j th time samples on the road m , y and $\bar{y}$ represent the set of y_m^j and $\widehat{y}_m^j$, respectively, $\bar{y}$ represents the average value of y , and N is the number of nodes on the road. For the two evaluation indicators of RMSE and MAE, the smaller the error value, the better the prediction effect of the model, while the larger the value of accuracy, VAR, and R^2 , the better the prediction effect.

Long- and short-term prediction

We divided the benchmark model into two groups for comparison: traditional algorithm and deep learning algorithm. Figure 8 shows the comparison of the prediction performance of the KR-STGNN model and baseline methods under different prediction horizons. The prediction horizons are set as 15, 30, 45, and 60 min. Table 2 provides the specific results predicted by each method on different evaluation metrics, where * indicates that the prediction performance is very inferior and the predicted value is small to be negligible. By comparing with different models, it can be found that the prediction effect of the proposed model in each prediction time step is better than that of most models, which proves its effectiveness in the traffic prediction task.

It can be found that the deep learning method has better prediction accuracy than traditional machine learning methods, such as SVR and ARIMA. For example, under the prediction task with a prediction horizon of 15 min, the RMSE error of KR-STGNN model is reduced by about 0.258 and 3.2106% compared with SVR and ARIMA, respectively,

Fig. 8 Trend of prediction results with increasing prediction steps under different methods. **a** RMSE. **b** MAE

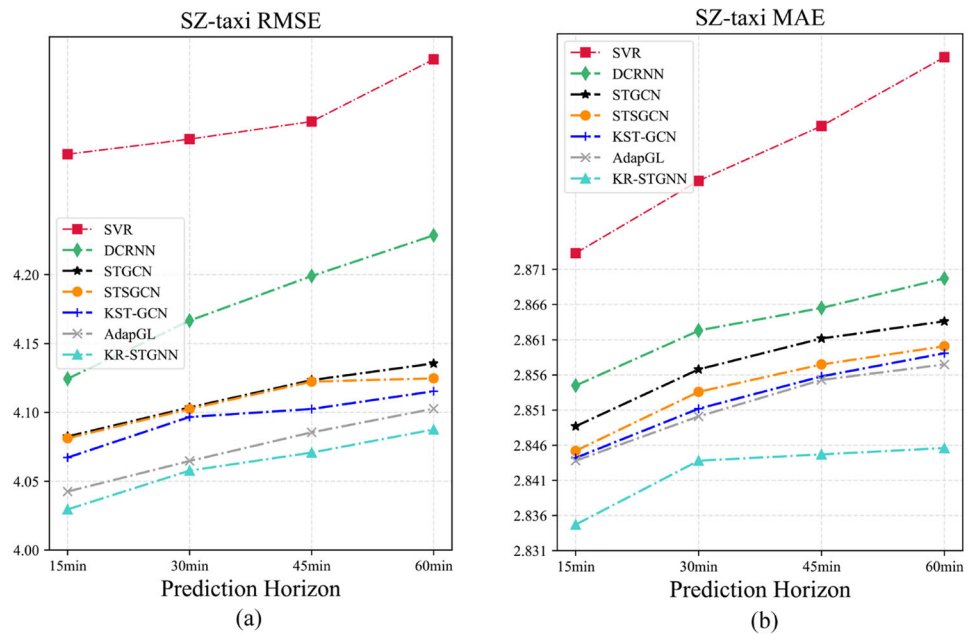


Table 2 Prediction results of KR-STGNN and other baseline methods on SZ-taxi data set

Time (min)	Metric	SVR	ARIMA	DCRNN	STGCN	STSGCN	KST-GCN	AdapGL	KR-STGNN
15	RMSE	4.2875	7.2401	4.1245	4.0825	4.0812	4.0672	4.0425	4.0295
	MAE	2.8733	4.6689	2.8579	2.8568	2.8452	2.8447	2.8338	2.8442
	Accuracy	0.7012	0.4463	0.7109	0.7116	0.7115	0.7120	0.7129	0.7247
	R^2	0.8107	*	0.8215	0.8266	0.8265	0.8270	0.8278	0.8306
	VAR	0.8112	0.0034	0.8215	0.8265	0.8265	0.8270	0.8287	0.8306
30	RMSE	4.2984	6.7902	4.1665	4.1036	4.1025	4.0967	4.0647	4.0577
	MAE	2.8836	4.6751	2.8598	2.8570	2.8536	2.8512	2.8440	2.8501
	Accuracy	0.7009	0.3856	0.7089	0.7010	0.7101	0.7112	0.7115	0.7232
	R^2	0.8100	*	0.8208	0.8252	0.8255	0.8258	0.8268	0.8287
	VAR	0.8107	0.0021	0.8205	0.8253	0.8255	0.8259	0.8270	0.8288
45	RMSE	4.3112	6.7862	4.1989	4.1235	4.1224	4.1025	4.0855	4.0708
	MAE	2.8914	4.6761	2.8612	2.8589	2.8575	2.8558	2.8553	2.8547
	Accuracy	0.6989	0.3789	0.7074	0.7008	0.7056	0.7107	0.7109	0.7218
	R^2	0.8085	*	0.8184	0.8216	0.8236	0.8248	0.8255	0.8269
	VAR	0.8076	0.0112	0.8189	0.8216	0.8237	0.8249	0.8253	0.8268
60	RMSE	4.3564	6.7789	4.2285	4.1356	4.1248	4.1154	4.1027	4.0876
	MAE	2.9012	4.6789	2.8699	2.8693	2.8631	2.8602	2.8589	2.8556
	Accuracy	0.6971	0.3756	0.7026	0.7004	0.7023	0.7101	0.7102	0.7206
	R^2	0.8075	*	0.8175	0.8211	0.8224	0.8228	0.8241	0.8255
	VAR	0.8069	0.0103	0.8174	0.8211	0.8224	0.8228	0.8240	0.8252

The values in the rightmost column are bolded to highlight that our results are better than most other models, and the two values in the penultimate column are bolded because our results are not better than those of other models

and the accuracy is increased by about 0.0235 and 0.2784%. This is mainly due to the difficulty of traditional machine learning methods in dealing with complex non-stationary time series data. It can be found that compared with DCRNN, STGCN, STSGCN, and AdapGL, the RMSE of KR-STGNN is reduced by 0.095, 0.053, 0.0517, and 0.013%, respectively.

In addition, the accuracy is improved by 0.0138, 0.0131, 0.0132, and 0.0118%, respectively, which is because these models do not consider the external influence factors that can affect the traffic situation, and do not jointly capture spatial-temporal features. Compared with KST-GCN, the RMSE is reduced by 0.0377%, and the accuracy is increased

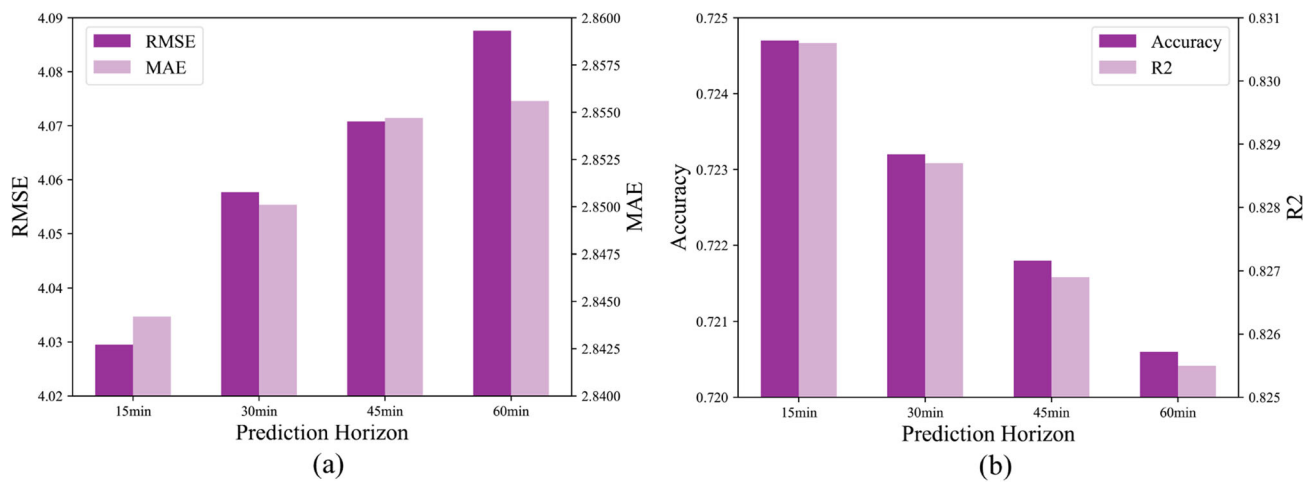


Fig. 9 Trends of each evaluation index of KR-STGNN at different prediction steps. **a** RMSE and MAE. **b** Accuracy and R^2

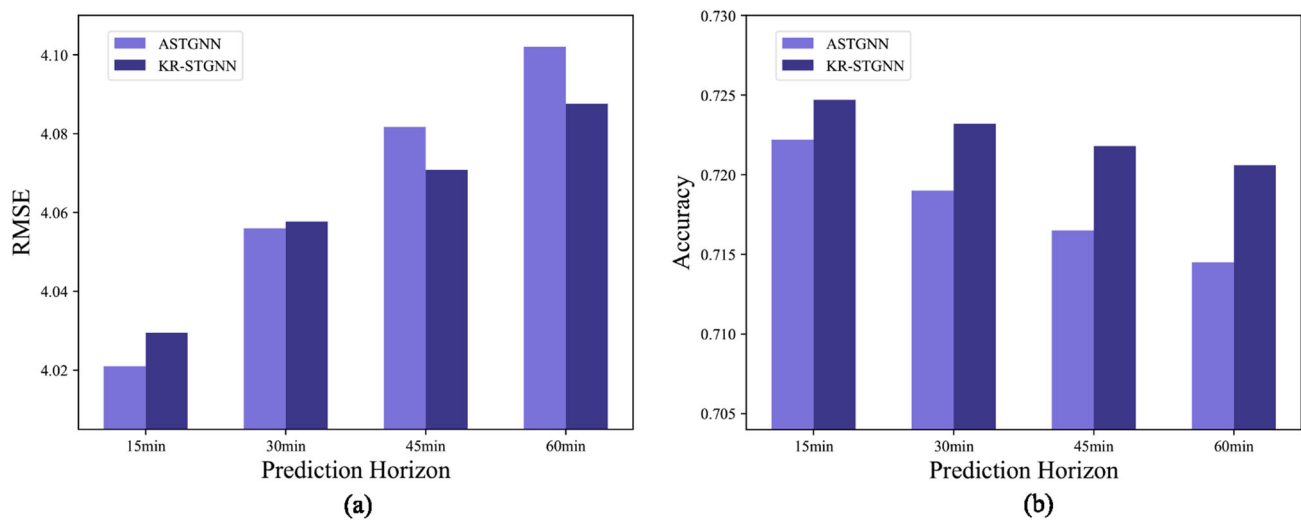


Fig. 10 Performance comparison with KR-STGNN and AST-STGNN. **a** RMSE. **b** Accuracy

by 0.0127%, because KST-GCN does not consider how to update traffic features adaptively and dynamically.

It can also be seen from Table 2 that KR-STGNN has relatively the best prediction effect in the short-term prediction task, and the prediction results gradually become worse as the prediction step increases. Based on the prediction horizon of 15 min, the RMSE of 30, 45, and 60 min increases by 0.0282, 0.0413 and 0.0581%, respectively, and the accuracy decreases by 0.0015, 0.0029 and 0.0041%, respectively. By visualizing it through Fig. 9, it can be seen that as the prediction horizon decreases, the prediction effect improves.

Analysis of knowledge representation significance

In this section, we demonstrate the significance of utilizing KRL method in traffic prediction. We introduce a model named ASTGNN, which concatenates external factors (static

and dynamic) with traffic features without utilizing KRL. We then generate an augmented matrices [31], which is subsequently fed into GFFM and ST-FSCM for prediction. To evaluate the performance, we conduct experiments across different prediction horizons and compare ASTGNN with our proposed model KR-STGNN. The results shown in Fig. 10 indicate that KR-STGNN outperforms ASTGNN in terms of accuracy, both for short-term and long-term predictions. In particular, KR-STGNN demonstrates improved accuracy compared to ASTGNN in short-term predictions, with the accuracy of 0.7247 and 0.7232 for 15 and 30 min forecasts, respectively. Furthermore, for long-term predictions of 45 and 60 min, KR-STGNN achieves the accuracy of 0.7218 and 0.7206, respectively, exhibiting improvement in performance. In addition, as the prediction horizon expands, the RMSE of ASTGNN gradually increases and eventually surpasses KR-STGNN by about 1.44% in the long-term

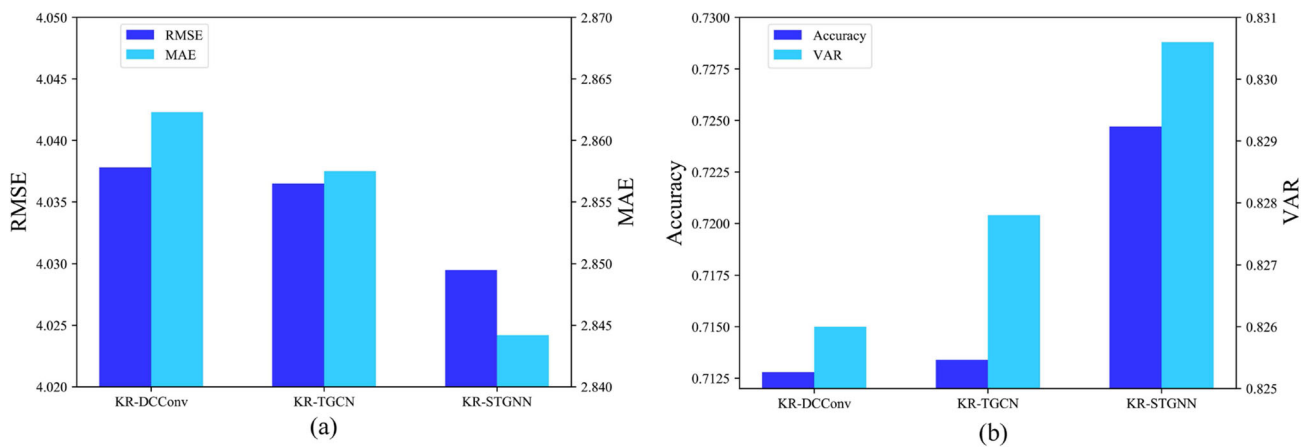


Fig. 11 Performance comparison with variant models. **a** RMSE and MAE. **b** Accuracy and VAR

prediction task of 60 min. It can also be observed that the performance gap between the two models gradually widens as the prediction horizon expands. The above results validate the indispensability of utilizing KRL method to account for external factors, thereby significantly enhancing the effectiveness of traffic prediction.

Ablation study

We conduct ablation experiments to analyze the impact of the GFFM and the ST-FSCM in the proposed model on the traffic prediction task. In the experiment, the model proposed in this paper is compared with two models, which are named KR-TGCN and KR-DCCConv, respectively. The former is a spatial–temporal graph convolutional network model after GFFM module, in which, in addition to using GCN to capture temporal correlation, the matrix multiplication operation in the traditional GRU is replaced by GCN operation to obtain spatial–temporal correlation. The latter is a model that simply concatenates the representation vector and traffic features obtained by the knowledge embedding method and inputs them into the spatial–temporal feature synchronization capture module. The results are shown in Fig. 11, and it can be seen that the KR-STGNN model proposed in this paper has better performance in all indicators than KR-TGCN and KR-DCCConv, which proves the effectiveness of the method of using causal dilated convolution to replace the fully connected layer in GRU and the ST-FSCM in the proposed model.

Perturbation analysis

In the real world, noise inevitably exists in the process of sensor data collection, and the ability to resist noise disturbance reflects the effectiveness and perturbation of the model. To test the effect of noise on KR-STGNN, we add two kinds

of noise to the original urban traffic data: Gaussian noise and Poisson noise. The added random noise follows Gaussian distribution $N \in (0, \sigma^2)$, $\sigma \in (0.2, 0.4, 0.6, 0.8, 1.2)$ and Poisson distribution $P(\lambda)$, $\lambda \in (1, 2, 4, 8, 16)$, respectively. The experimental results after adding two types of noise, respectively, are shown in Fig. 12. From the results, we can conclude that the changes of each index after adding noise are small, indicating that the presence of noise does not significantly affect the performance of KR-STGNN, and KR-STGNN is robust to noise in the data.

Visualized analysis

On the basis of SZ-taxi data set, we visualize the prediction results of the proposed KR-STGNN model, and predict the traffic of 15, 30, 45, and 60 min in the future based on the historical data in a historical time, as shown in Figs. 13, 14, 15 and 16. For all prediction ranges, KR-STGNN can well capture the changing trend of traffic data, so that it can predict more accurately. Through the comparison of the four figures, it can be seen that the short-term prediction effect of KR-STGNN is better than that of long-term prediction, and the predicted value of short-term prediction is closer to the real value than that of long-term prediction, which may be due to the more complex spatial–temporal dependencies in long-term prediction. In addition, there is a certain deviation at the speed inflection point in the KR-STGNN prediction process, and the reason may be that the change of peak time is affected by factors, such as weather, POIs, or traffic accidents.

Conclusion

In this paper, we propose a traffic prediction model KR-STGNN based on knowledge representation learning of spatial–temporal graph convolution network. Our model

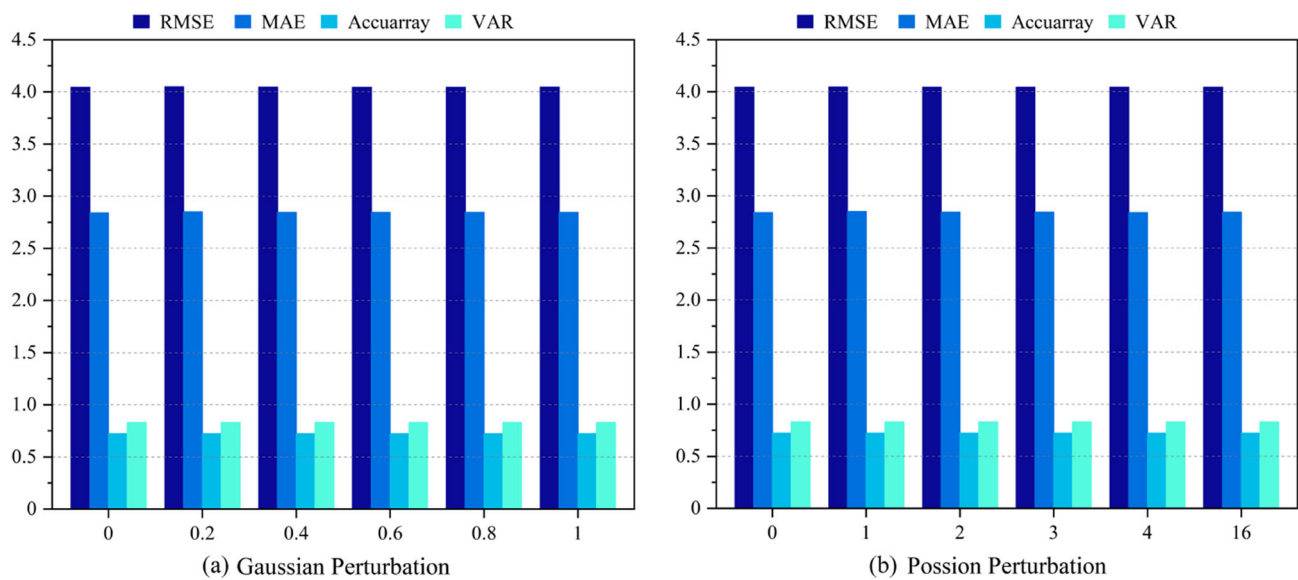
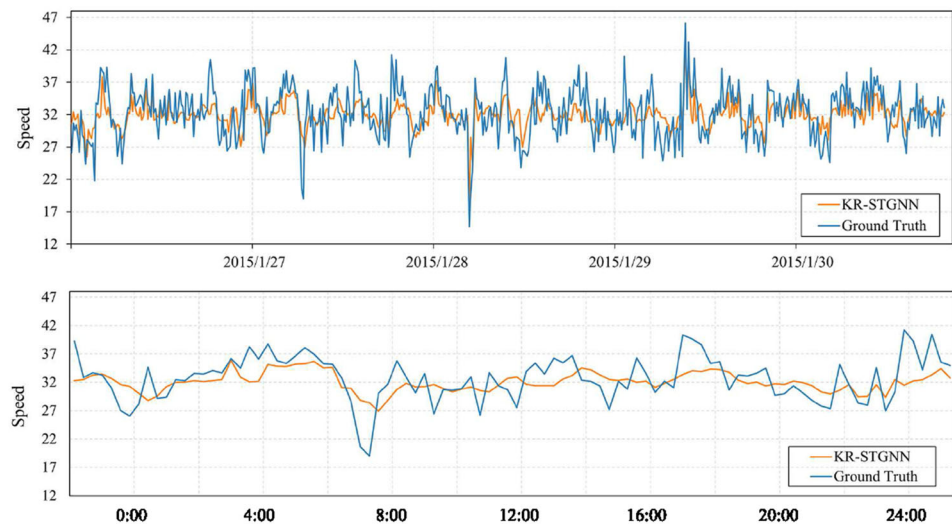


Fig. 12 Perturbation analysis of KR-STGNN. **a** Gaussian noise is added to the input data. **b** Poisson noise is added to the input data

Fig. 13 Visualization result with a prediction horizon of 15 min



attempts to solve the problem that traditional urban traffic forecasting methods ignore the correlation between traffic information and external factors, and fail to integrate the two types of information. It utilizes knowledge representation learning to extract external knowledge and adapts traffic features with GFFM module to account for the varying importance of different factors. In addition, ST-FSCM module effectively co-captures the spatial-temporal dependencies in the prediction task. The experimental results demonstrate that KR-STGNN outperforms the baseline methods at different prediction horizons, especially in short-term traffic flow prediction, which is 1.3% superior to the state-of-the-art baseline method and improves the accuracy by 1.18%. In addition, the ablation and perturbation analysis experiments

verify the effectiveness and robustness of KR-STGNN. Due to limitations in data collection, this study only considers external factors, such as POIs, weather information and road sections. Therefore, the expressiveness and performance of KR-STGNN is limited. In the future, if more road network topological feature data and other types of data (such as road length, number of lanes, events and social media data) can be obtained, it would be advantageous to enhance the model input and construct a more comprehensive knowledge graph. This augmentation would undoubtedly contribute to the improved prediction capabilities of KR-STGNN.

Fig. 14 Visualization result with a prediction horizon of 30 min

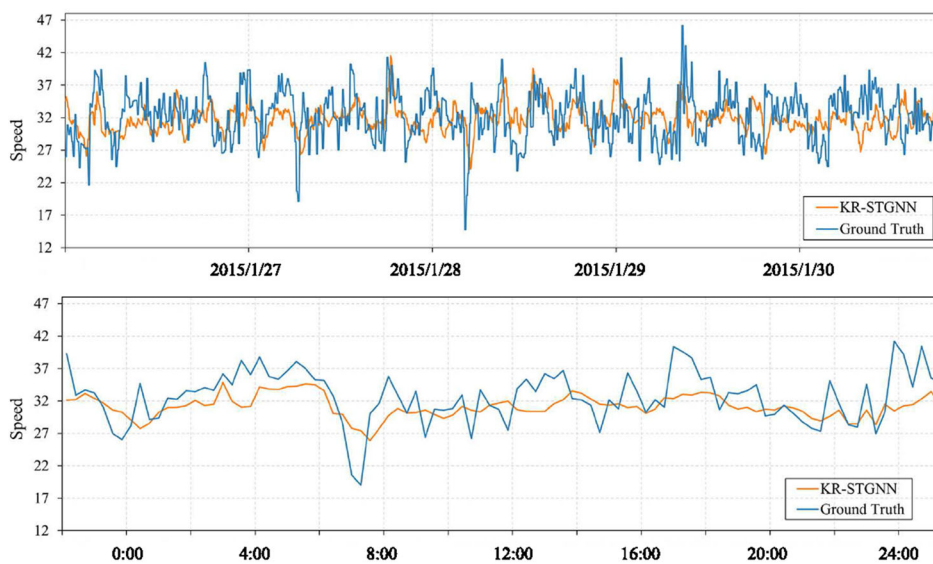


Fig. 15 Visualization result with a prediction horizon of 45 min

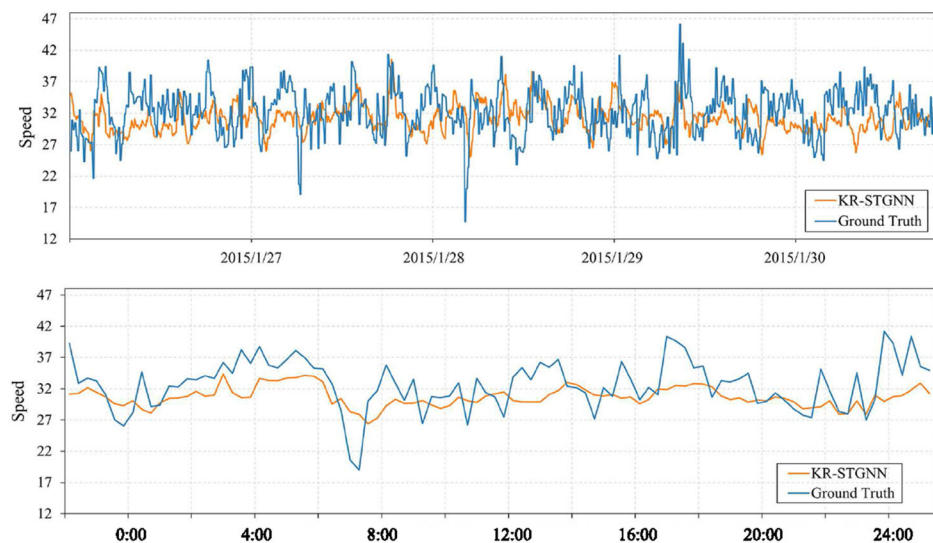
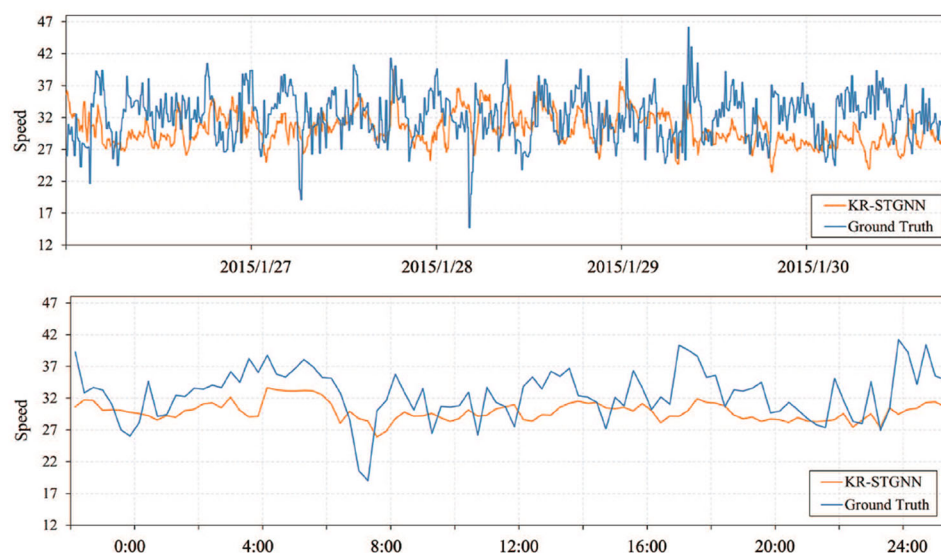


Fig. 16 Visualization result with a prediction horizon of 60 min



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Declarations

Conflict of interest Corresponding authors declare on behalf of all authors that there is no conflict of interest. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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